

DISHA THALWAR

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TECHNICAL SKILLS

- **Programming Languages:** C++, Java, Javascript, HTML, PHP, PowerShell and Python | **Database:** MySQL, AWS and Oracle
- **Tools:** NetBeans, XAMPP, Eclipse, Visual Studio, Anaconda, Jupyter Notebook, PyCharm, Sharepoint and Neuroph
- **Certifications:** Machine Learning, **Cybersecurity Professional**, **Digital Tagging and Agile Developer**
- **Security Tools:** Familiarity with industry-standard security tools, including **BurpSuite**, **Tcpdump**, **Metasploit**, **Sysinternals**, **Echo mirage**, **Snort**, **Wireshark**, **Postman**, **Nmap** and **Nessus**
- **Security:** **Application Security**, **Penetration Testing**, **OWASP**, **Malware Analysis**, **Threat Hunting**, **Vulnerability Assessment**, **Vulnerability Scanning** and **Risk Assessment**
- **Machine Learning:** **Deep Learning**, **Neural Network**, **Keras**, **Tensorflow**, **Optimization Techniques** and **Transfer Learning**

EDUCATION

University of Texas at Arlington, Texas, USA

May 2025

Pursuing Master of Science in Computer Science

(GPA 3.8/4.0)

Relevant Coursework: Design and Analysis of Algorithms, Data Analysis and Modeling Techniques, Data Mining, Machine Learning, Software Engineering, Artificial Intelligence, Cloud Computing and Big Data, Distributed Systems.

Savitribai Phule Pune University, Pune, India

May 2020

Secured a Bachelor of Engineering in Computer Engineering with *Distinction*

(CGPA 8.03/10)

Relevant Coursework: Data structures and Algorithms, Database Management Systems, Software Development, Artificial Intelligence and Robotics, Data Mining and warehousing, Cloud Computing, Cyber Security, Object-Oriented Programming.

PROFESSIONAL EXPERIENCE

Infosys, Pune, India | Security Engineer-IVS (Cybersecurity)

June 2021-July 2023

- Provided cybersecurity expertise and conducted 100+ comprehensive vulnerability assessments, penetration testing initiatives for Deutsche Bank, resulting in strengthened security measures and mitigation of critical vulnerabilities.
- Spearheaded a cross-functional team to execute 60+ in-depth Vulnerability Assessment and Penetration Testing (VAPT) engagements on web applications and Google Cloud-hosted environments, uncovering and mitigating critical security vulnerabilities to strengthen system resilience.
- Proficient in performing Host-Based Audits across Linux, Windows servers, Oracle Databases, and ensuring compliance with CIS Benchmark guidelines and industry standards such as CVE, CVSS, HIPAA, CWE, PCI-DSS, NIST 800-53, ISO 27001.
- Collaborated with the clients to comprehend their needs before the release of an application and provide technical insight about the bugs and vulnerabilities present in the application.
- Delivered detailed reports and recommendations for remediation to both technical and non-technical audiences, resulting in a 20% increase in application reliability and client satisfaction.

High Energy Materials Research Laboratory (Defense Research and Development Organization) | Machine Learning Intern

- Developed and deployed 5 high-performance machine learning models to detect anomalous user behavior, enhancing attack prediction capabilities and fortifying system security.
- Achieved a 95% prediction accuracy, reducing false positives by 20%, and enhancing threat detection efficiency.
- Conducted extensive research on Artificial Intelligence, Deep Learning, and Big Data applications in cybersecurity, utilizing real-time datasets exceeding 10 million records, employing advanced data preprocessing techniques, and executing rigorous experiments.
- Earned recognition from a Senior Scientist for exceptional contributions, leading to sponsorship for the final-year project

ACADEMIC PROJECTS AND RESEARCH PAPER

Intrusion Detection using R-CNN Algorithm | Python, AI, Anaconda, Jupyter Notebook, GitHub

- Modeled a system (Sponsored by High Energy Materials Research Laboratory (Defense Research and Development Organization)), with R-CNN algorithm for detecting intrusion in the network, resulting in 15% increase in accuracy.
- Trained an R-CNN model to classify network intrusions, including DDoS attacks, malware activity, and unauthorized access, achieving a 92% detection accuracy by analyzing datasets and developing defense mechanisms.

Convolutional Neural Networks for Image Classification | Python, AI, Anaconda, Jupyter Notebook, GitHub

- Efficiently deployed the VGG-16 convolutional neural network architecture on the CIFAR-100 dataset.
- Introduced a novel approach by applying the VGG-16 model to image classification, in contrast to previous studies from the research paper that focused on CNN architectures like GoogLeNet, AlexNet, and ResNet50.
- Demonstrated the superior effectiveness of the VGG-16 model by achieving 86% accuracy, surpassing the average performance of other CNN architectures (AlexNet: 36.12%, GoogLeNet: 71.67%, ResNet50: 78.10%).

Distributed Linear Regression | Apache Spark, Scala

- Implemented distributed matrix computations, and their integration with Breeze for pseudo-inverse calculations.
- Compared algorithm performance on the Boston Housing dataset, analyzing execution time and accuracy for different implementations.

RPC-based Communication System | Ubuntu, Python

- Engineered automated file synchronization with helper threads, creating a Dropbox-like storage service, and built a computation server with synchronous/asynchronous RPC for addition and sorting.